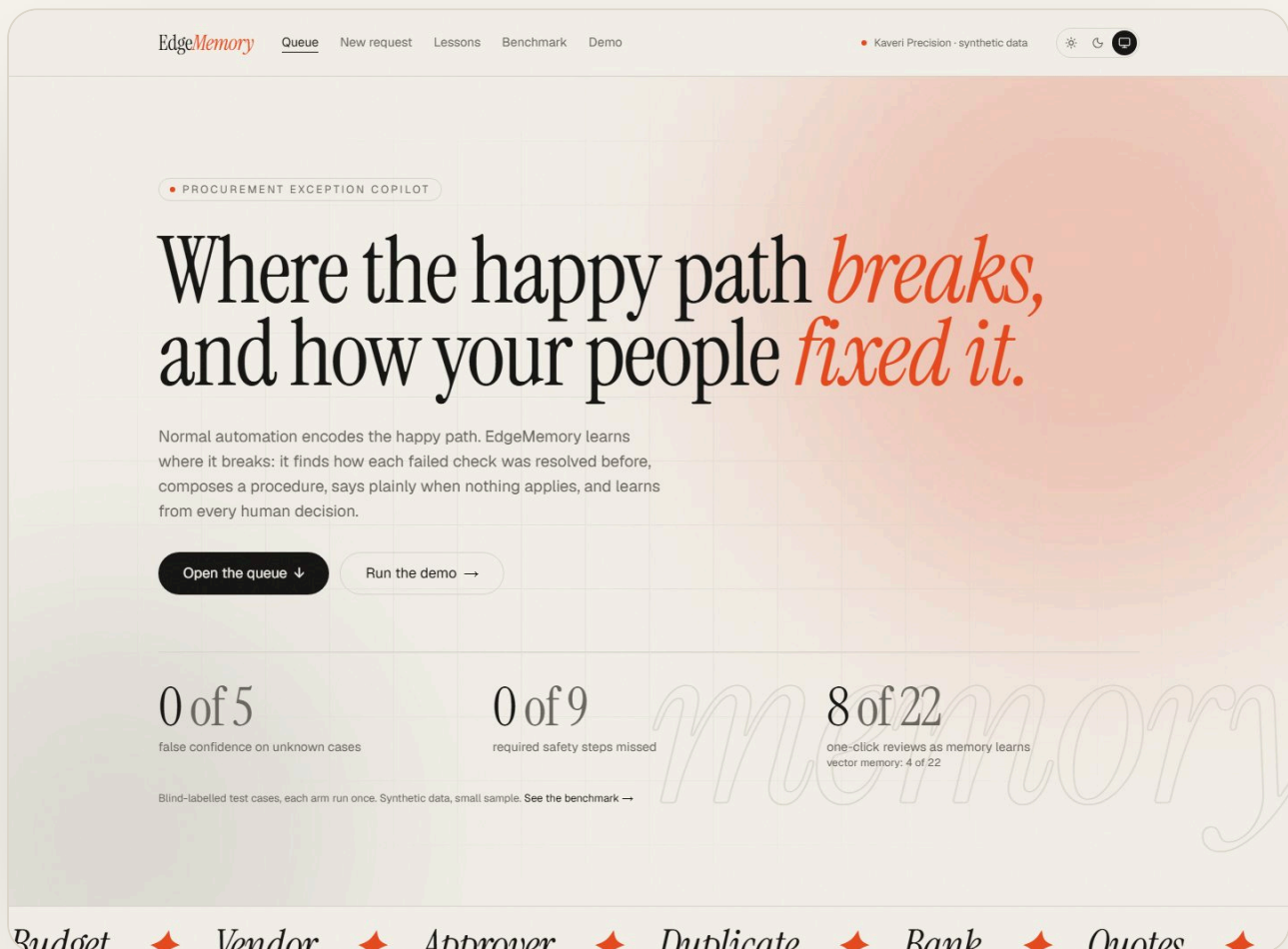


The user *guide*

What EdgeMemory is, how it works, how to set it up and use it, and what to expect: in plain language, with pictures of the real screens.



Inside this *guide*

01	What is EdgeMemory?	the idea in two minutes
02	How it works	from request to decision
03	Where to use it	who it's for, where it fits
04	Set it up	install and start, step by step
05	A tour of the screens	what each page shows
06	Guided walkthrough	run the demo and check it works
07	What to expect	speed, results, limits
08	Troubleshooting	when something goes wrong
09	Glossary	the words, explained

The short version

EdgeMemory helps the people who handle purchase requests that **break the normal rules**. It remembers how similar problems were solved before, suggests the steps to take, says clearly when it has never seen a problem, and learns from every decision a person makes. **It never approves anything on its own. A human always decides.**

A memory for the *exceptions*

Most purchase requests are routine. The ones that aren't (an expired certificate, a budget overrun, an approver on leave) take most of the time, and the know-how to fix them lives in people's heads and old emails.

The problem

In a company like *Kaveri Precision Components* (a made-up mid-size manufacturer in Hyderabad), every purchase request goes through a set of checks: Is there budget? Is the vendor in good standing? Is the right person available to approve? Is this a duplicate? Did the vendor's bank details change? Were enough quotes collected?

When a check fails, the request lands with a procurement analyst. They have to work out what to do: who to ask, what paperwork to collect, whether to hold the order. Usually someone solved the same problem before, but finding out how takes time. When several checks fail at once, the answer can be different from simply doing each fix.

What EdgeMemory does

- **Finds the precedents.** For every check that failed, it looks up how people resolved that same problem before.
- **Checks they really apply.** A precedent only counts if the situation matches: same problem, same cause, similar amount and category, and no policy change since.
- **Suggests a procedure.** It combines the lessons into one list of steps, and every step links back to the real past cases behind it.
- **Says "I don't know" when it doesn't.** If nothing matches, it sends the case to a person, with what is known and what isn't.
- **Learns from you.** Every approve, edit, reject or resolve is remembered, so the next similar case is easier.

What it never does

- ✗ It never approves, rejects or pays anything by itself. It only recommends; a person decides.
- ✗ It never releases a payment.
- ✗ It never removes a safety step (such as "hold the payment until the new bank account is confirmed by phone").
- ✗ It never gives a confident answer to a problem it has never seen.

About the data

Everything you see in this guide and in the app (the company, people, vendors, amounts and past cases) is **synthetic**: made up for the demo. Only synthetic data is ever sent to the AI services this build uses.

From request to *decision*

Every request goes through the same five steps. The first four are automatic and take about 10–20 seconds. The fifth is yours.

STEP 1

Read

An AI reads the request, emails and attachments (even messy ones like "1.9L" or Hinglish) and pulls out the facts.

STEP 2

Check

Six fixed rules run: budget, vendor, approver, duplicate, bank, quotes. Plain code, no AI.

STEP 3

Recall

For each failed check, memory finds past resolutions, and a verifier confirms each one really applies.

STEP 4

Compose

The lessons are combined into one procedure. A safety floor adds any step that must never be skipped.

STEP 5

You decide

Approve, confirm, edit, reject or resolve. The decision is remembered for next time.

Four ideas that make it different

IDEA 1

One problem at a time

A request with three failed checks isn't "one big unknown". EdgeMemory looks for a precedent for **each** check separately, so it can say "two of these three are familiar, one is new".

IDEA 2

Problems can change each other

Sometimes two problems together need a different fix. Example: an approver on leave *plus* a budget overrun means the delegate can't approve; it must go one level up. EdgeMemory learns these combinations from past decisions; nobody programs them in.

IDEA 3

Lessons grow up

A lesson backed by one case is **tentative**: you check each step. Once several cases agree it becomes **established** and can be approved in one click. If a reviewer overrules a lesson, it is revised. If policy changes, old lessons are flagged as out of date.

IDEA 4

Honest about the unknown

If any failed check has no real precedent, the case goes to a person, with what is known and what isn't. The goal is **zero false confidence**: never a confident answer that wasn't earned.

The five kinds of case

Kind	What it means	How you review it
Normal	Every check passed.	One click
Known	One problem, and a precedent that applies directly.	One click if the lesson is established; otherwise step by step
Generalized	One problem, and a precedent that applies with some differences (listed for you).	Step by step
Composed	Several problems, each with a precedent. The procedure combines them, adjusted for how they interact.	Step by step
Unknown	At least one problem has no real precedent.	Goes to a person to resolve

The safety floor

Some steps can never be removed by any lesson. For example: if a vendor's bank details changed, EdgeMemory always adds *verify the bank account by phone* and *hold the payment*. If the requester is related to the vendor, it always adds *declare the conflict of interest*. If a vendor's tax (GST) registration is cancelled, it always holds the order.

Behind the scenes: memory is **Hindsight** (by Vectorize). It stores each resolved case, merges similar cases into lessons, and keeps track of which cases each lesson came from, which is how every step can cite real case numbers.

Who it's for, and *where it fits*

Who it's for

- **Procurement operations analysts:** the people who handle requests that fail the standard checks. This is who EdgeMemory is built for.
- **Approvers and team leads:** they see a clear procedure with the reasons and past cases behind it, instead of a long email chain.
- **Finance controllers and auditors:** every recommendation, citation, safety step and human decision is logged with a timestamp and can be exported.

Where it fits in your day

EdgeMemory sits next to the approvals queue you already have. Requests can come from a monitored approvals inbox or from an export of your existing system. For each exception, it adds a recommendation: the steps, a citation for every step, and what isn't covered.

How a team would start

- 1 **Load your history.** Export the last year of resolved exceptions (what failed, what was done, why) into a simple spreadsheet format. EdgeMemory learns its first lessons from that. The demo starts exactly this way.
- 2 **Shadow mode.** EdgeMemory recommends; analysts keep working as usual and compare.
- 3 **Step-by-step review.** After about two weeks, analysts start ticking each recommended step.
- 4 **One-click review, lesson by lesson.** As lessons become established, those cases can be approved in one click. Unknown cases always go to a person; that never switches off.

Other places the same idea could help

The approach (check each problem separately, learn how problems combine, learn from every human decision) could apply to other rule-heavy approval work, such as accounts-payable exceptions, IT change approvals or insurance claims. **This build has only been built and tested for procurement.**

Using real company data

This hackathon build runs on free AI tiers and must only process synthetic data. A real deployment would use a paid, zero-data-retention AI setup (or the company's own AI provider) and a private memory store. The code is built so only one file changes for that.

Install and *start*

About 15 minutes the first time. You need two programs, three free keys, and two terminal windows.

What you need

Item	Details
Python 3.11 or newer	Runs the engine and the API (the "back end").
Node.js 20 or newer	Runs the website (the "front end").
Groq API key (free)	The main AI model, <code>openai/gpt-oss-120b</code> . From console.groq.com .
Gemini API key (free)	Backup AI models, used only if Groq fails. From Google AI Studio.
Hindsight Cloud key	The memory. From Vectorize's Hindsight Cloud.

Step by step (Windows PowerShell)

- 1 **Open the project folder** in PowerShell.

```
cd D:\EdgeMemory
```

- 2 **Add your keys.** Copy the example settings file, then open `.env` in a text editor and paste your three keys between the quotes.

```
copy .env.example .env
```

- 3 **Install the back end** (once).

```
python -m venv .venv
.venv\Scripts\python -m pip install -r requirements.txt
```

- 4 **Install the website** (once).

```
cd web
npm install
cd ..
```

- 5 **Start the API** and leave this window open.

```
.venv\Scripts\python -m uvicorn api.main:app --port 8000
```

- 6 **Start the website** in a second PowerShell window and leave it open too.

```
cd D:\EdgeMemory\web
npm run dev
```

7 Open the app at `http://localhost:3000`. Go to **Demo**, click **Reset**, then **Seed**. Seeding loads 37 past cases into memory and takes about 3 minutes.

On macOS or Linux, or if you have "make"

The same steps are shortcuts: `make setup` installs, `make api` and `make web` start the two parts, and `make test` runs the automated tests. Use `.venv/bin/python` in place of `.venv\Scripts\python`.

How to tell it's running

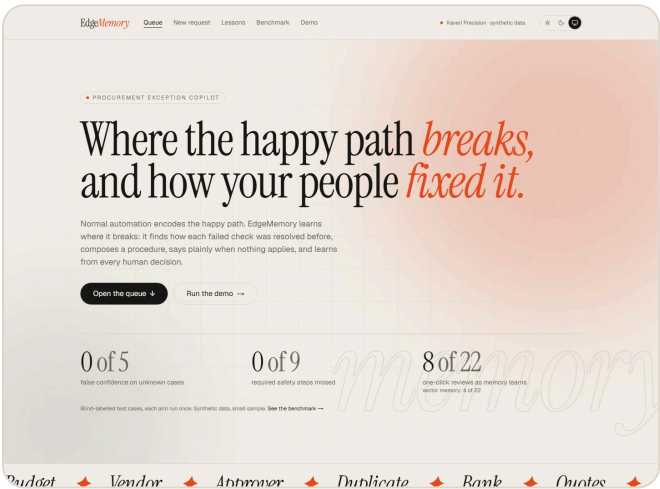
The website opens at `localhost:3000`. The **Demo** page shows how much free AI quota is left and the API response time. The API's own page is at `localhost:8000/docs`. To stop, press **Ctrl+C** in each window.

What you're *looking at*

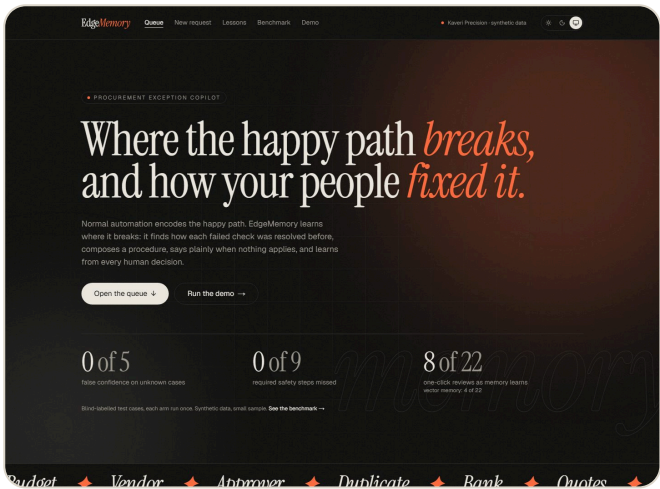
Six pages, all in the top menu. The switch at the top right changes between light, dark, and your computer's setting.

Home and the queue

The home page explains EdgeMemory in one line and shows headline results from the benchmark. The numbers come straight from the scored test runs. Scroll down for how it works, then **the queue**: open requests grouped by kind, most urgent first (cases that need a person at the top). Click any row to open it.



Light theme.



Dark theme.

The case page: seven stages

Opening a request shows how EdgeMemory reached its recommendation, in seven numbered stages. In demo mode they appear one after another, so you can follow the reasoning.

#	Stage	What it tells you
1	The request	Vendor, amount, cost centre, and the original text. Important phrases (like a family link) are highlighted.
2	Failed checks	Which of the six checks failed, and exactly why.
3	How each problem was resolved before	For each failed check: whether a precedent applies, the past case numbers, and how mature the lesson is.
4	The plain combination	The steps you'd get by simply adding up each problem's lesson.
5	What changes together	Steps removed or added because these problems happened <i>together</i> , each with the past cases that justify it.
6	Safety floor	Any must-have steps added by the safety rules.
7	Recommended procedure	The final steps, each with its past cases and a Why this step? button, plus the decision buttons.

← Queue

PR-DEMO-HOOK

Composed

established

Step-by-step review

Replay reveal U

3 of 3 problems have precedents · 1 interaction precedent applies · payment held until the bank callback

01 The request

Vendor
Srinivasa Castings (V-102)

Amount
₹4,80,000 (2-10L)

Cost centre · category · urgency
CC-MACH · indirect MRQ · line down

Justification: Line 3 band hai since 2 am. Need replacement spindle housings for the line 3 machining centre today. 3 quotes attached, Srinivasa is L1. Cost centre is over budget but line cannot stay down.

Email thread: From: maintenance@kaveri

Line 3 stopped at 02:05, maintenance log MNT-5541. Housings cracked, no spares in stock.

From: purchase@kaveri

Iyer sir is on leave till 2 Oct. Raising the PR anyway.

02 Failed checks (3)

Budget (overrun)

₹4,80,000 exceeds the ₹0 left in cost centre CC-MACH (FY 2026-27, allocated ₹80,00,000); 6.0% over budget

Approver (approver on leave)

Approver E-201 (plant_head, CC-MACH) is on leave 2026-09-21 to 2026-10-02; delegate E-205 up to ₹10,00,000, valid 2026-04-01 to 2027-03-31

Bank (bank details changed)

bank details changed on 2026-09-12 (12 days before the request)

The top of a case: the request (stage 1) and its three failed checks (stage 2).

03

How each problem was resolved before

Budget

precedent applies

established (6 cases)

Lesson panel →

Cases PR-2026-0205, PR-2026-0344, PR-2026-0421, PR-2026-0498, PR-2026-0561, PR-2026-0655. Budget overrun with line down matches emergency overrun precedent; same band and category.

Approver

applies with differences

established (6 cases)

Lesson panel →

Cases PR-2026-0301, PR-2026-0391, PR-2026-0460, PR-2026-0548, PR-2026-0720, PR-2026-0758. Approver on leave with amount within delegate limit for indirect_mro. Differences: no cited case in the same amount band and category.

Bank

applies with differences

established (6 cases)

Lesson panel →

Cases PR-2026-0118, PR-2026-0233, PR-2026-0309, PR-2026-0455, PR-2026-0597, PR-2026-0672. Bank change cause matches, but amount band differs from precedent. Differences: amount band 2-10L vs precedent up to 2L.

Precedents where these problems happened together

approver+budget

precedent applies

cases PR-2026-0360, PR-2026-0515, PR-2026-0620

bank+budget

no precedent

cases PR-2026-0312, PR-2026-0484, PR-2026-0690

Stage 3: each problem's precedents. Budget matches directly; approver and bank apply with differences, which are listed. A past case of the same pair of problems (approver + budget) also applies.

04

The plain combination of single-check lessons

Verify bank callback

Hold payment

Hold po

Attach line down evidence

Route-delegate

Add controller signoff

Allow budget overrun

Expedite po

05

What changes when these problems occur together

Route delegate

Removed because the request includes a budget overrun; process must route to the next-level approver to prevent processing stalls. (PR-2026-0360, PR-2026-0620)

Hold po

conflict with EXPEDITE_PO (precedent PR-2026-0360)

+ Route next level approver

Required because the primary approver is absent and the request includes a budget overrun; delegates are not authorized to approve overruns. (PR-2026-0360, PR-2026-0620)

06

Safety floor

The safety floor holds without additions. Payment stays held until the bank callback.

07

Recommended procedure

☐ Verify bank callback

PR-2026-0309

Why this step?

☐ Hold payment

PR-2026-0309

Why this step?

☐ Attach line down evidence

PR-2026-0655

Why this step?

☐ Allow budget overrun

PR-2026-0655

Why this step?

☐ Add controller signoff

PR-2026-0655

Why this step?

☐ Route next level approver

PR-2026-0360, PR-2026-0620

Why this step?

☐ Expedite po

PR-2026-0205, PR-2026-0344, PR-2026-0421, PR-2026-0498, PR-2026-0561, PR-2026-0655

Why this step?

dropped_uncited_step:EXPEDITE_PO

restored_removed_step:HOLD_PO

restored_removed_step:EXPEDITE_PO

Confirm 0 of 7 steps

Edit

Reject

Stages 4 to 7. The plain combination would route to the delegate, but because the budget is also over, the procedure routes to the next-level approver instead, citing the past cases where that happened. Payment stays held until the bank callback.

Your decision buttons

Button	When you see it	What happens
Approve	One-click cases (established lessons)	Accepts the procedure as it is.
Confirm	Step-by-step cases	Tick each step first, then confirm.
Resolve	Unknown cases	You write what should be done. This becomes a brand-new lesson.
Edit	Always	Change the steps. Your version is remembered as a correction.
Reject	Always	Overrule the lesson for this situation. EdgeMemory revises the lesson for similar cases.

Why this step?

Click it next to any step to see the past cases behind it, what memory recalled, and the verifier's verdict. Nothing is recommended without a real past case you can check.

Lessons: what memory has learned

Pick a check (for example `step:vendor`) to see the current lesson in plain words, the cases it was built from, and **how it changed**: removed words are struck through in red, new words are highlighted in green.

MEMORY

What it has *learned*

Per failed check and per combination: the current consolidated lesson, the cases behind it, and how it changed as people decided.

step:vendor

step:budget

step:approver

step:duplicate

step:bank

step:quotes

combo:approver+budget

combo:bank+budget

signal:related_party

Current lesson (Hindsight observation)

For vendors in good standing with a recently expired ISO 9001 certificate, the procurement process is modified to allow continued activity while awaiting renewal; the procedure involves step codes REQUEST_RENEWED_CERT, KEEP_VENDOR_ACTIVE, and HOLD_PO, as applied in cases PR-DEMO-EI-01, PR-DEMO-EI-02, PR-DEMO-R001, and PR-DEMO-R002.

Consolidated from 4 cases: PR-DEMO-EI-01 PR-DEMO-EI-02 PR-DEMO-R001 PR-DEMO-R002

How this lesson changed (3 revisions)

Before: For vendors in good standing with a recently expired ISO 9001 certificate, the procurement process is modified to allow continued activity while awaiting renewal; the procedure involves step codes REQUEST_RENEWED_CERT, KEEP_VENDOR_ACTIVE, and HOLD_PO, as applied in cases PR-DEMO-EI-01, PR-DEMO-EI-02, and ~~PR-DEMO-R001~~.

After: For vendors in good standing with a recently expired ISO 9001 certificate, the procurement process is modified to allow continued activity while awaiting renewal; the procedure involves step codes REQUEST_RENEWED_CERT, KEEP_VENDOR_ACTIVE, and HOLD_PO, as applied in cases PR-DEMO-EI-01, PR-DEMO-EI-02, **PR-DEMO-R001**, and **PR-DEMO-R002**.

Cases in the ledger under step:vendor

PR-DEMO-EI-01	experience	2026-09-24	REQUEST_RENEWED_CERT, KEEP_VENDOR_ACTIVE, HOLD_PO
PR-DEMO-EI-02	confirmation	2026-10-01	HOLD_PO, REQUEST_RENEWED_CERT, KEEP_VENDOR_ACTIVE
PR-DEMO-R001	confirmation	2026-10-10	REQUEST_RENEWED_CERT, KEEP_VENDOR_ACTIVE, HOLD_PO
PR-DEMO-R002	confirmation	2026-10-17	REQUEST_RENEWED_CERT, KEEP_VENDOR_ACTIVE, HOLD_PO

The certificate lesson after four decisions. It started from one case; the latest revision added two more cases.

New request

Type or paste a request the way people really write them. The example is pre-filled. Press **Submit**; after 10–60 seconds the new case opens with the staged reveal.

INTAKE

A new *request*

Write it the way people do: amounts like 1.9L, Hinglish, a messy email thread.
EdgeMemory extracts the facts, runs the checks and recalls precedents.
Synthetic data only; the example is pre-filled.

Purchase request

Requester

E-402

Cost centre

CC-MACH

Quotes attached

1

Vendor

Nizam Pumps

Amount (as written)

1.9L

Justification

Line 3 band hai, coolant pump seized. Need pump assemblies today. Budget over but line cannot stay down.

Email thread

From: maintenance@kaveri
Line 3 stopped at 02:10, log MNT-5530.

Attachment text (certificate OCR)

Submit

The new-request form, with the pre-filled example.

Benchmark: does it hold up?

The results of testing EdgeMemory against simpler approaches on 24 made-up test cases whose correct answers were written down and locked before testing. The highlighted column is EdgeMemory. See section 07 for what the numbers mean.

Test cases 24, each arm run once

Measure	A no memory	B plain RAG	C · EdgeMemory Hindsight	D same pipeline, vector store	E C without interaction learning
False confidence on unknown cases (lower is better)	3 of 5	1 of 5	0 of 5	0 of 5	0 of 5
Required safety steps missing (lower is better)	7 of 9	4 of 9	0 of 9	0 of 9	0 of 9
Case type identified correctly (higher is better)	11 of 24	15 of 24	21 of 24	21 of 24	21 of 24
Procedure accuracy (F1, 0 to 1) (higher is better)	0.472	0.843	0.798	0.797	0.758
Interaction procedures exactly right (higher is better)	0 of 8	0 of 8	3 of 8	3 of 8	0 of 8
New combinations flagged (higher is better)	0 of 4	0 of 4	4 of 4	4 of 4	4 of 4
Outdated pre-July lesson used (lower is better)	0 of 3	0 of 3	0 of 3	2 of 3	3 of 3
Correct after a reviewer override (higher is better)	1 of 3	3 of 3	0 of 3	0 of 3	0 of 3
Sent to a human unnecessarily (lower is better)	10 of 17	3 of 17	3 of 17	1 of 17	3 of 17
Reviewer effort per 10 cases (lower is better)	25.4	37.5	28.3	39.6	35.8
Estimated minutes per 10 exception cases (lower is better)	270	206.4	189.5	166.4	204.5
LLM tokens per case (lower is better)	2,300	4,374	3,699	6,931	3,699

Where EdgeMemory is behind: plain RAG (B) handles cases after a reviewer override better, and D sends fewer cases to a human. B and A also give confident answers on unknown cases and miss required safety steps; C, D and E never do. Source: report-test-20260928T222947.json.

The results table. Lower is better for mistakes; higher is better for correct answers.



The learning curve: the test cases replayed in date order, with each decision remembered. The orange line is EdgeMemory with Hindsight; the grey line is the same system with plain search memory.

Demo controls

Everything needed to run the scripted demo: the eight steps (each opens its case), the demo clock, **Reset** and **Seed** for the demo memory, and how much free AI quota is left today.

STAGE

Demo*controls*

Reset and seed the demo memory, then run the scripted beats in order. Each beat opens its case with the staged reveal.

Demo script (SPEC §18)

Run in order after Reset and Seed. Each beat sets the demo date and opens the case in demo mode.

1. Hook: ₹4.8L, line 3 down, 6% over budget, approver on leave, bank changed 12 Sep

C1, C2

Run

open

2. First exception: ₹60,000, vendor certificate expired 6 days ago (no precedent yet)

C4 → C3

Run

open

3. One week later: a different vendor, the same problem (tentative, 1 precedent)

C3

Run

open

4. Replay 3 weeks: two more certificate cases resolved the same way

C3

Run

5. The next certificate case: established (3 cases), one click

C3

Run

open

6. Split shipment from a vendor with a past duplicate-invoice incident: the reviewer rejects

C3

Run

open

7. A similar case follows the revised lesson

C3

Run

open

8. Honest limit: the vendor's owner is the requester's brother-in-law

C4

Run

open

Demo clock

10/22/2026

Set

New requests are dated to this day.

Demo bank

Reset

Seed (37 memories, ~3 min)

Replay 3 weeks

LLM quota left (free tiers)

openai/gpt-oss-120b: 997 requests, 1,65,759 tokens left (last 24 h)

gemini-3.8-flash: 0 requests left (Pacific day 2026-09-28)

gemini-3.7-flash: 4 requests left (Pacific day 2026-09-28)

API round trip: 10 ms

The Demo page. Run the steps in order after Reset and Seed.

Run the demo, *step by step*

This is the best way to test EdgeMemory and see it learn. It takes about 10 minutes. Go to **Demo**, click **Reset** then **Seed**, and wait about 3 minutes. Then click **Run** on each step in order.

1 The hook: three problems at once

The request	₹4.8 lakh, production line 3 is down, 6% over budget, the approver is on leave, and the vendor's bank details changed 12 days ago.
You'll see	Composed , step-by-step. All three problems have precedents, and a past case of <i>approver away + budget over</i> changes the plan: it goes to the next-level approver, not the delegate. Payment is held until the bank is verified by phone.
Do	Nothing yet. Come back to it at the end (pictured in section 05).

2 A problem it has never seen

The request	₹60,000 to a vendor whose quality certificate expired 6 days ago. Memory has no certificate cases yet.
You'll see	Unknown and Escalate to a human, with "No verified precedent".
Do	Click Resolve. The form is pre-filled with a sensible answer. Save it. It becomes the first certificate lesson.

← Queue

PR-DEMO-EI-01

Unknown

Escalate to a human

Replay reveal ↻

0 of 1 problem has precedents · 1 problem escalated to a human

01 The request

Vendor

Deccan Bearings (V-101)

Amount

₹60,000 (up to 2L)

Cost centre · category · urgency

CC-ASSY · Direct Material · normal

Justification: Bearing housings for the assembly line, regular item. Deccan has supplied for 2 yrs, never late.

Email thread: From: sales@deccanbearings

Our ISO 9001 renewal audit is done; the new certificate will reach you next week.

Attachment: ISO 9001:2015 ... Deccan Bearings ... Valid until: 18-09-2026

02 Failed checks (1)

Vendor (cert expired)

Deccan Bearings (V-101): ISO 9001 certificate expired 2026-09-18 (6 days before the request)

03 How each problem was resolved before

Vendor

no precedent

Lesson panel →

No verified precedent. This problem goes to a human, with what is known and what isn't.

Step 2: no precedent, so EdgeMemory says so and sends it to a person.

3 One week later: the same problem again

The request	A different vendor, the same kind of expired certificate.
You'll see	It now recognises the problem and cites PR-DEMO-E1-01, the case you resolved in step 2. The lesson is tentative (one case), so the review is step by step.
Do	Tick the three steps and click Confirm.

[← Queue](#)

PR-DEMO-E1-02

Known tentative Step-by-step review

Replay reveal to

1 of 1 problem has precedents

01 The request

Vendor

Charminar Fasteners (V-108)

Amount

₹95,000 (up to 2L)

Cost centre · category · urgency

CC-ASSY · Direct Material · normal

Justification: Stainless fasteners that go into the pump housings we ship to customers. Charminar is our regular vendor, good quality.

Email thread: From: charminar.fasteners

Our ISO certificate renewal is in process; the surveillance audit was completed last week.

Attachment: ISO 9001 ... Charminar Fasteners ... Expiry: 19/09/2026

02 Failed checks (1)

Vendor (cert expired)

Charminar Fasteners (V-108): ISO 9001 certificate expired 2026-09-19 (12 days before the request)

03 How each problem was resolved before

Vendor

precedent applies tentative (1 case)

Lesson panel →

Cases PR-DEMO-E1-01. All conditions match precedent; same check, cause, amount band, category, and vendor standing.

Step 3: learned from one decision, so the lesson is tentative and each step is checked.

4 Three weeks pass

What happens	Two more certificate cases are resolved the same way in the background, and the demo clock moves forward.
Do	Optional: open Lessons → <code>step:vendor</code> to see the lesson now backed by more cases, with the change highlighted.

5 The lesson is established

You'll see **Known**, established (4 cases), and a single Approve button: one-click review.
Do Click Approve.

← Queue

PR-DEMO-EI-03

Known established One-click review

Replay reveal ↻

1 of 1 problem has precedents

Step 5: the same problem, now established.

07 Recommended procedure

Hold po	PR-DEMO-EI-01, PR-DEMO-EI-02, PR-DEMO-R001, PR-DEMO-R002	Why this step?
Request renewed cert	PR-DEMO-EI-01, PR-DEMO-EI-02, PR-DEMO-R001, PR-DEMO-R002	Why this step?
Keep vendor active	PR-DEMO-EI-01, PR-DEMO-EI-02, PR-DEMO-R001, PR-DEMO-R002	Why this step?

dropped_uncited_step:VERIFY_BANK_CALLBACK dropped_uncited_step:HOLD_PAYMENT

Approve Edit Reject

The procedure cites all four past cases and can be approved in one click.

6 You overrule a lesson

The request A "split shipment" from a vendor that once sent a duplicate invoice.
You'll see The usual split-shipment lesson: link it to the main order and allow the split delivery.
Do Click Reject. The form is pre-filled: this vendor's history means the claim can't be taken on trust. Save it. EdgeMemory records this as an override for this situation.

7 It follows your correction

The request	Another split shipment from the same vendor.
You'll see	The old steps are crossed out, and the procedure is now Reject request and Escalate to audit , citing PR-DEMO-E4-01, your decision in step 6. The original lesson still holds for other vendors.
Do	Confirm.

04

The plain combination of single-check lessons

Link master po

Allow split delivery

05

What changes when these problems occur together

~~Link master po~~ — Prohibited due to the vendor's documented history of duplicate invoicing. (PR-DEMO-E4-01, PR-2026-0266, PR-2026-0402, PR-2026-0477, PR-2026-0569, PR-2026-0648)

~~Allow split delivery~~ — Prohibited due to the vendor's documented history of duplicate invoicing. (PR-DEMO-E4-01, PR-2026-0266, PR-2026-0402, PR-2026-0477, PR-2026-0569, PR-2026-0648)

+ **Reject request** — The request must be rejected because the vendor has a confirmed history of duplicate invoicing. (PR-DEMO-E4-01)

+ **Escalate audit** — The case must be escalated to internal audit for a thorough review of the vendor's recent invoicing behavior. (PR-DEMO-E4-01)

06

Safety floor

The safety floor holds without additions.

07

Recommended procedure

☐ **Reject request** PR-DEMO-E4-01 Why this step?

☐ **Escalate audit** PR-DEMO-E4-01 Why this step?

Confirm 0 of 2 steps

Edit

Reject

Step 7: the reviewer's correction from step 6 changes the recommendation for a similar case.

8 An honest limit

The request	The vendor's owner signs an email as the requester's brother-in-law.
You'll see	The family link is highlighted. Unknown : there is no precedent for a related party, so it goes to a person. The safety floor still adds Declare conflict of interest.
Do	Nothing. This shows EdgeMemory refusing to guess.

← Queue

PR-DEMO-U1-01

Unknown Escalate to a human Replay reveal

0 of 1 problem has precedents · 1 step added by safety floor · 1 problem escalated to a human

01 The request

Vendor	Amount	Cost centre · category · urgency
Warangal Wires (V-111)	₹1,40,000 (up to 2L)	CC-QA · Services · normal

Justification: Wire harness testing service for the QA lab, 3 month contract.

Email thread: From: owner@warangalwires
Hi Gopal, as discussed at the family function, sending our rates. Regards, Srikanth (your brother-in-law)

From: g.rao@kaveri
Thanks, raising the request.

signal: related party (email thread)

Step 8: the highlighted phrase is why the related-party check failed.

06 Safety floor

Added by safety floor: Declare conflict of interest

07 Recommended procedure

Declare conflict of interest Added by safety floor Why this step?

F3:no_one_click uncovered:related_party

Resolve Edit Reject

No precedent, so no guess: only the safety-floor step, and the case goes to a person.

It's working if...

- ✓ Steps 2 → 3 → 5 show the same problem going **unknown** → **tentative** → **established**.
- ✓ Step 7 follows your step-6 rejection instead of the old lesson.
- ✓ Steps 2 and 8 go to a person instead of guessing.
- ✓ The hook (step 1) and step 8 keep their safety-floor steps.
- ✓ Every step's **Why this step?** shows real case numbers.

Small differences are normal

The AI isn't perfectly repeatable, so a class or a step may occasionally differ from the pictures (for example "Known" instead of "Generalized"). Steps 2 and 8 must always go to a person, and step 7 must always recommend rejecting and escalating. Before recording a demo, click **Reset** and **Seed** again, because running the steps changes the memory.

Speed, results and *limits*

Speed

Action	Typical time
A new request (read, check, recall, compose)	10–20 seconds (up to about 60 for a new request)
Saving a decision (remember and merge into lessons)	6–10 seconds
Seeding the demo memory	About 3 minutes
The whole 8-step demo, including reset and seed	About 5 minutes

Results on the test cases

EdgeMemory (C) was compared with: **A**, an AI with the policy but no memory; **B**, an AI with plain document search; and **D**, the same EdgeMemory pipeline with plain search memory instead of Hindsight. 24 made-up test cases; each approach ran once; the correct answers were locked before scoring.

Measure	A	B	C · EdgeMemory	D
Confident answer on a case it should have escalated (lower is better)	3 of 5	1 of 5	0 of 5	0 of 5
Required safety steps missing (lower is better)	7 of 9	4 of 9	0 of 9	0 of 9
Kind of case identified correctly	11 of 24	15 of 24	21 of 24	21 of 24
Combined problems handled exactly right	0 of 8	0 of 8	3 of 8	3 of 8
Used an out-of-date lesson after a policy change (lower is better)	0 of 3	0 of 3	0 of 3	2 of 3
Right answer after a reviewer override	1 of 3	3 of 3	0 of 3	0 of 3

Learning: when the test cases were replayed in date order with every decision remembered, EdgeMemory with Hindsight reached a one-click review on **8 of 22** exception cases, against **4 of 22** with plain search memory, and gave **no** false-confidence answers along the way.

Where it's weaker

Plain search (B) handled the three "after a reviewer override" cases better. EdgeMemory sent two to a person and gave one an empty procedure. That problem has since been fixed and checked on the demo, but the published test numbers are from before the fix and weren't re-run. EdgeMemory also sends a few cases to a person that didn't strictly need it.

Free-tier limits

- The main AI (Groq, free tier) allows about **200,000 tokens a day**, refilling gradually. The full demo uses about 16,000.

- The backup models (Gemini, free tier) allow **20 requests a day each**.
- The Demo page shows what's left. If it runs out, cases are sent to a person with "Memory unavailable" instead of a guess.

Good to know

- All data is synthetic, and there is one domain (procurement) and a small sample of cases.
- Time savings shown on the benchmark page are estimates from stated assumptions, not measurements.
- The test answers were written by an AI labeller working blind, not checked by a person.

When something *goes wrong*

What you see	What to do
The website won't open at localhost:3000	Check the website window is still running (<code>npm run dev</code>). Start it again if it stopped.
Pages load but show "Couldn't load..." errors	The API isn't running. Start it in its own window (step 5 of setup) and refresh.
Every case says "Memory unavailable, escalating to a human"	The memory service or the AI can't be reached. Check the internet connection and the keys in <code>.env</code> , then look at the API window for errors.
A step is stuck on "Running..." for more than a minute	The free AI quota is probably used up for now. Check the quota on the Demo page; it refills gradually.
A case cites itself, or the demo looks different from this guide	A step was run twice without resetting. Click Reset then Seed on the Demo page and run the steps in order.
A page shows an error after the computer was busy for a long time	Stop the website window (Ctrl+C) and start it again with <code>npm run dev</code> .
The animations are distracting	Turn on "reduce motion" in your computer's settings; the app switches its animations off.

Check the engine itself

`.venv\Scripts\python -m pytest -q` runs 170 automated tests offline in a few seconds. They should all pass.

The words, *explained*

Word	Meaning
Purchase request (PR)	A request to buy something. The six checks run on it.
Purchase order (PO)	The document that commits the company to buy. "Hold PO" means don't issue it yet.
Exception	A request that failed at least one check and needs a person's attention.
Precedent	A past case where people resolved the same kind of problem. Every recommended step cites one.
Lesson	What memory has learned from several precedents, in plain words.
Tentative / established	A lesson from one case is tentative (check each step); one backed by several agreeing cases is established (one click).
Override	A reviewer rejected or changed what a lesson said for a specific situation. Later similar cases follow the reviewer.
Interaction	Two or more problems happening together, where the right fix differs from doing each fix separately.
Safety floor	Steps no lesson can remove, such as holding payment until a changed bank account is confirmed by phone.
False confidence	Giving a confident procedure for a problem with no real precedent. The target is zero.
Escalate	Send the case to a person to decide.
Cost centre	A department's budget line (for example CC-MACH, machining).
Delegate	The person allowed to approve while the usual approver is away, up to a set amount.
Line down	A production line has stopped, which makes the request urgent.
Sole source	Only one approved vendor can supply the item, so the usual three quotes aren't possible.
GST	India's Goods and Services Tax. A cancelled GST registration is a serious vendor problem.
Lakh (L)	Indian unit: 1 lakh = ₹1,00,000. So ₹4.8L = ₹4,80,000.
Hindsight	The memory service (by Vectorize) that stores cases, merges them into lessons, and tracks where each lesson came from.
Synthetic data	Made-up data. Kaveri Precision Components and everything about it is fictional.

EdgeMemory recommends; a human always decides. It never releases a payment. Built for HackwithHyderabad 3.0 with Hindsight memory. All data in this guide is synthetic.